

```
pip install scikit-learn
```

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Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.1.0)
```

import

```
from sklearn.linear_model import LogisticRegression
```

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from sklearn.model_selection import train_test_split
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```
from sklearn.metrics import accuracy_score, classification_report
```

```
from sklearn.datasets import load_iris, load_digits
```

1.import 2.prepare data 3.split 4.train 5.predict

```
import pandas as pd
import numpy as np
```

```
#creating the data set
data={
    'hours_studied':[1,2,3,4,5,6],
    'pass_fail':['yes','yes','yes','no','no','no']
}
df=pd.DataFrame(data)
print(df)

#seperate
x=df[['hours_studied']]
y=df[['pass_fail']]

#split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
print(f'train:{len(x_train)} rows |Test:{len(x_test)} rows')

#create and train the model
model=LogisticRegression()
model.fit(x_train,y_train)
print('model is sucessfully trained')

#evlauate on the set
y_pred=model.predict(x_test)
accuracy_score(y_test,y_pred)

#predict
print(f'accuracy{accuracy:.2%}')
```

```
hours_studied pass_fail
0            1      yes
1            2      yes
2            3      yes
3            4      no
4            5      no
5            6      no
train:4 rows |Test:2 rows
model is sucessfully trained
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:1408: DataConversionWarning: A column-vector y was
  y = column_or_1d(y, warn=True)
-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_4923/619125590.py in <cell line: 0>()
    25
    26 #predict
--> 27 print(f'accuracy{accuracy:.2%}')
```

NameError: name 'accuracy' is not defined

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